ADNI Knowledge Graph
Detailed Node & Relationship Report

What every node means, what every relationship does,
why CN maps to 'No diagnosis', and the full 4-phase roadmap

Project: ADNI Knowledge Graph for Alzheimer's Disease Research
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Current State: Phase 1 complete (Steps 17-20) | Phase 2 next (Steps 21-23)

1. Graph Overview (Live Data from Neo4j)

All numbers below are **live counts** queried directly from the running Neo4j 5.24.2 database. The graph contains data from **108 ADNI CSV tables** spanning demographics, cognitive assessments, CSF biomarkers, neuroimaging, genetic profiles, and family history for Alzheimer's disease research.

Metric	Count
Total Nodes	407,422
Total Relationships	1,391,812
Unique Node Labels	37
Unique Relationship Types	52
Patients (ADNI participants)	2,638
Visits (clinical encounters)	30,267
Diagnosis nodes	25,946
Cognitive Assessments	65,345
Biomarker measurements	12,008
Medical Images (MRI/PET)	88,769
Family Members	97,797
Brain Regions	12
OntologyConcept nodes	51

Table 1: Live graph statistics from Neo4j.

2. Every Node Type Explained

Below is a detailed explanation of every node label in the graph. For each node, we show what it represents in the context of Alzheimer's disease research, what properties it carries, and how many instances exist.

Patient (2,638 nodes)

Each Patient node represents one **ADNI study participant**. ADNI enrolls people across the cognitive spectrum: healthy controls (CN), those with mild cognitive impairment (MCI), and Alzheimer's disease (AD) patients. These are the hub nodes — everything connects to a Patient.

> **Key properties:** ptid (unique ID like '011_S_0002'), rid, gender (M/F), education_years, rdf_type='ncit:C16960' (NCI Thesaurus: Research Subject)

Visit (30,267 nodes)

Each Visit represents a **clinical encounter** where a patient came to the hospital for assessments. ADNI has baseline visits (bl), 6-month (m06), 12-month (m12), etc. Each visit can produce multiple assessments, biomarker measurements, and imaging scans.

> **Key properties:** visit_id (unique: ptid+viscode), patient_id, viscode (bl/m06/m12...), rdf_type='ncit:C159705' (NCI Thesaurus: Clinical Visit)

Diagnosis (25,946 nodes)

Each Diagnosis records **what cognitive status** a patient had at a specific visit. The three main codes are: **CN** (Cognitively Normal = healthy control), **MCI** (Mild Cognitive Impairment = early decline), **AD** (Alzheimer's Disease). Distribution: CN=13,526 **|** MCI=9,582 **|** AD=2,838.

> **Key properties:** diagnosis_code (CN/MCI/AD), diagnosis_text, snomed_code, icd10_code, icd10_label, rdf_type, visit_id, patient_id

CognitiveAssessment (65,345 nodes)

Each node is one **cognitive test score** for a patient at a visit. Tests include MMSE (Mini-Mental State), CDR (Clinical Dementia Rating), ADAS-Cog (Alzheimer's scale), MoCA, FAQ, and Logical Memory. Higher MMSE = better; higher CDR = worse.

> **Key properties:** test_name (MMSE/CDR/ADAS-Cog/MoCA/FAQ/Logical Memory), total_score, loinc_code (e.g. 72106-8 for MMSE), loinc_label, visit_id

Biomarker (12,008 nodes)

Each node is a **CSF biomarker measurement**. The key biomarkers are: **ABETA42** (amyloid-beta 42, drops when AD starts), **TAU** (total tau, rises with neurodegeneration), **PTAU181** (phosphorylated tau, rises with tangle formation). These three biomarkers form the **ATN framework** (Amyloid-Tau-Neurodegeneration).

> **Key properties:** analyte (ABETA42/TAU/PTAU181), value, unit, biomarker_type='CSF', loinc_code (e.g. 13967-5 for Abeta42), specimen_type, visit_id

ImageNode (88,769 nodes)

Each node represents a **medical image** (MRI or PET scan) processed by the pipeline. The IEEE Big Data 2025 paper validated 100% pixel preservation across DICOM -> TIFF/PNG conversion. Each image has metadata indexed in Elasticsearch for millisecond retrieval.

> **Key properties:** image_id, patient_id, modality (MR/PT), image_type, file_paths, quality_metrics, processing_status

FamilyMember (97,797 nodes)

Each node represents a **relative of a patient** (parent, sibling, child) with information about whether they had Alzheimer's or dementia. Family history of AD is a major risk factor.

Key properties: member_id, patient_id, relationship_type (parent/sibling), gender, ad_status

BrainRegion (12 nodes)

Each node is an **anatomical brain region** from FreeSurfer parcellation. The hippocampus is the most important — its volume loss is an early marker of AD. Each region carries a UBERON ontology code for global identification.

Key properties: name (Hippocampus, Cerebral Cortex, Ventricles, ...), uberon_code (e.g. UBERON:0002421 for Hippocampus), uberon_label

OntologyConcept (51 nodes)

These are the **semantic backbone** of the Knowledge Graph. Each node represents a formal concept from a biomedical ontology (SNOMED-CT, LOINC, UBERON, HPO, ICD-10). They enable machine reasoning and interoperability with external knowledge bases.

Key properties: uri (globally unique, e.g. snomed:26929004), code, label, source_ontology

DiseaseStage (5 nodes)

Five ordered stages of cognitive progression: **CN** (normal) -> **SMC** (subjective memory concern) -> **EMCI** (early MCI) -> **LMCI** (late MCI) -> **AD** (Alzheimer's). These stages model the Jack et al. (2010) biomarker cascade.

Key properties: stage_id (CN/SMC/EMCI/LMCI/AD), name, order (1-5)

ATNProfile (2,638 nodes)

One per patient. Records the patient's **ATN classification** based on biomarkers: A+ (amyloid positive), T+ (tau positive), N+ (neurodegeneration positive). A+T+N+ = full AD pathology; A-T-N- = healthy.

Key properties: patient_id, a_status (+/-), t_status (+/-), n_status (+/-)

3. Key Relationships Explained

The graph has **52 relationship types** with **1,391,812 total edges**. Below are the most important ones, grouped by function.

Relationship	From -> To	Count	URI	Meaning
HAS_COGNITIVE_ASSESSMENT	Visit -> CogAssess	161,274	ro:RO_0002234	Visit produced this test score
PROGRESSED_TO	DxStage -> DxStage	119,561	ro:RO_0002411	Patient progressed to next stage
SUPPORTS_DIAGNOSIS	CogAssess -> Dx	115,879	ro:RO_0000091	Test score supports this diagnosis
HAS_DIAGNOSIS	Visit -> Diagnosis	114,122	ro:RO_0000091	Diagnosis at visit
MAPS_TO	Data -> OntConcept	100,770	skos:exactMatch	Links data to ontology concept
HAS_IMAGE	Visit -> ImageNode	80,764	ro:RO_0000056	Imaging at visit
HAS_FAMILY_MEMBER	Patient -> FamilyMbr	74,613	ro:RO_0002351	Patient's relative
HAS_BIOMARKER	Visit -> Biomarker	42,720	ro:RO_0000056	Biomarker at visit
CLASSIFIED_AS	Diagnosis -> ICD-10	25,946	skos:closeMatch	ICD-10 disease classification
HAS_VISIT	Patient -> Visit	23,773	ro:RO_0000056	Patient attended visit
IS_A	OntConcept -> Parent	27	rdfs:subClassOf	Taxonomic hierarchy

Table 2: Key relationship types with live counts and semantic URIs.

4. Why CN Maps to "No Diagnosis" in ICD-10

This section directly addresses Prof. Turhan's question: *'What do CN nodes represent in the ICD-10 relationships? Why does the middle node say No diagnosis?'*

CN = Cognitively Normal. These are **healthy control participants** in the ADNI study. They are enrolled specifically because they do NOT have any cognitive disease. ADNI needs healthy controls as a baseline comparison group for the MCI and AD patients.

ICD-10 code Z03.89 means *'Encounter for observation for other suspected diseases and conditions ruled out'*. In medical coding, when a patient is examined and found to be healthy, you do not leave the ICD-10 field empty — you use a Z-code to indicate 'we checked, and they do not have a disease'. This is standard medical practice.

The three diagnosis categories in our graph and their ICD-10 mappings:

Diagnosis Code	Full Name	ICD-10 Code	ICD-10 Label	SNOMED Code	Count
CN	Cognitively Normal	Z03.89	No diagnosis or condition	17621005	13,526
MCI	Mild Cognitive Impairment	F06.7	Mild cognitive disorder	386806002	9,582
AD	Alzheimer's Disease	G30.9	Alzheimer disease, unspecified	26929004	2,838

Table 3: Diagnosis distribution with ICD-10 and SNOMED-CT mappings.

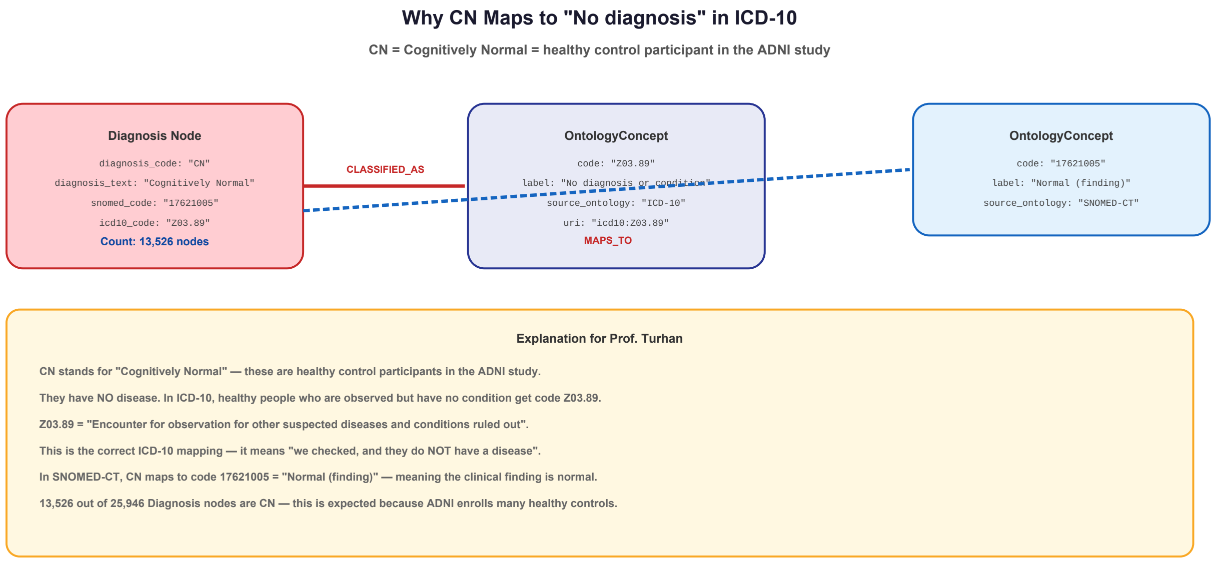

Figure 1: How CN Diagnosis nodes connect to ICD-10 'No diagnosis' and SNOMED 'Normal finding'.

In summary: CN mapping to 'No diagnosis or condition' is **medically correct**. It means these participants were assessed and found to be cognitively healthy. The ICD-10 Z03.89 code is specifically designed for this scenario in clinical coding.

5. The Ontology Layer — Where Is the Knowledge Base Connection?

This addresses Prof. Turhan's question: *'Where is the Knowledge Base connection?'*

The **OntologyConcept nodes** ARE the knowledge base connection. They link our clinical data to global biomedical standards. Here is what each ontology provides:

- **SNOMED-CT (18 concepts):** The world's largest clinical terminology system. Our diagnoses are grounded in SNOMED: Alzheimer's = 26929004, MCI = 386806002. The IS_A hierarchy goes: Alzheimer's -> Dementia -> Neurodegenerative disorder -> Disorder of nervous system -> Disease.
- **LOINC (10 concepts):** The universal standard for lab tests and clinical measurements. Every cognitive test has a LOINC code: MMSE = 72106-8, CDR = 72172-0. Every CSF biomarker has one too: Abeta42 = 13967-5, Tau = 15201-7, pTau = 62731-6.
- **UBERON (14 concepts):** The cross-species anatomy ontology. Every brain region is identified: Hippocampus = UBERON:0002421. The IS_A hierarchy: Hippocampus -> Cerebral cortex -> Brain.
- **ICD-10 (5 concepts):** The WHO's International Classification of Diseases. Used for disease classification: G30.9 (AD), F06.7 (MCI), Z03.89 (CN). The IS_A hierarchy: G30.9 -> G30 (Alzheimer disease group).
- **HPO (5 concepts):** Human Phenotype Ontology. Maps to observable clinical features: Cognitive impairment, Dementia, Memory impairment.

How traversal works (real example from Neo4j):

Patient 002_S_0619 -> Visit -> Diagnosis (AD) -> CLASSIFIED_AS -> ICD-10 G30.9 (Alzheimer disease, unspecified) -> IS_A -> G30 (Alzheimer disease)

Same Diagnosis -> MAPS_TO -> SNOMED 26929004 (Alzheimer's disease) -> IS_A -> Dementia -> IS_A -> Neurodegenerative disorder -> IS_A -> Disorder of nervous system -> IS_A -> Disease

This semantic traversal was **impossible** in the LPG state. It is what makes this a Knowledge Graph.

6. Why Only In-Place Semantic Conversion? The Full Roadmap

This addresses Prof. Turhan's observation: *'In your document there is only in-place semantic conversion.'*

Yes, Phase 1 was deliberately limited to in-place semantic conversion. This is by design. The ADNI_KG_Design_Blueprint.pdf describes a **4-phase architecture**. We completed Phase 1 today (February 24). The remaining three phases will progressively add:

- **Phase 1 (DONE today):** Add the semantic backbone — ontology codes on existing nodes, OntologyConcept layer with IS_A hierarchies, MAPS_TO and CLASSIFIED_AS edges. This transforms the LPG into a KG **without rebuilding** — we keep all existing data intact.
- **Phase 2 (Steps 21-23, NEXT):** Extract a feature matrix from the KG (demographics, cognitive scores, CSF biomarkers, brain volumes, genetic data). Run **causal discovery algorithms** (PC, FCI, GES) to find which variables cause which. Create **CAUSES relationships** in the graph. Expected: Amyloid -> Tau -> Neurodegeneration -> Cognitive decline.
- **Phase 3 (Steps 24-26):** Connect to **AlzKB** (external Alzheimer's knowledge base with 118,902 entities) via SAME_AS edges. Validate discovered causal edges against known AD biology. Run DoWhy causal inference for effect estimation and refutation tests.
- **Phase 4 (Steps 27-28):** Generate final statistics, publication-quality figures, and thesis defense materials.

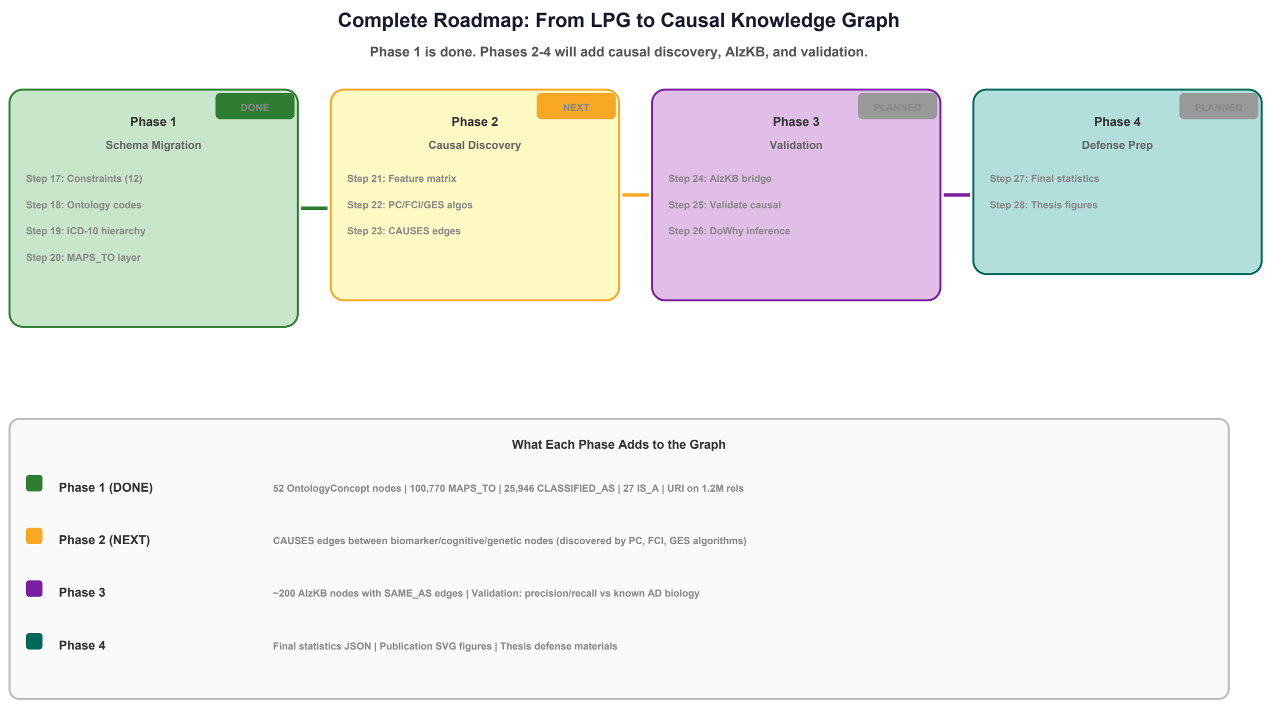

Figure 2: Complete 4-phase roadmap from LPG to Causal Knowledge Graph.

The in-place approach was chosen because: (1) Rebuilding 407K nodes would take days and risk data loss. (2) Prof. Turhan's feedback specifically asked for ontology grounding — Phase 1 delivers exactly that. (3) The existing data relationships (HAS_VISIT, HAS_DIAGNOSIS, etc.) are correct and do not need to change — they just needed semantic annotation.

7. Target Architecture (from Design Blueprint)

The final graph after all 4 phases will have **four layers**:

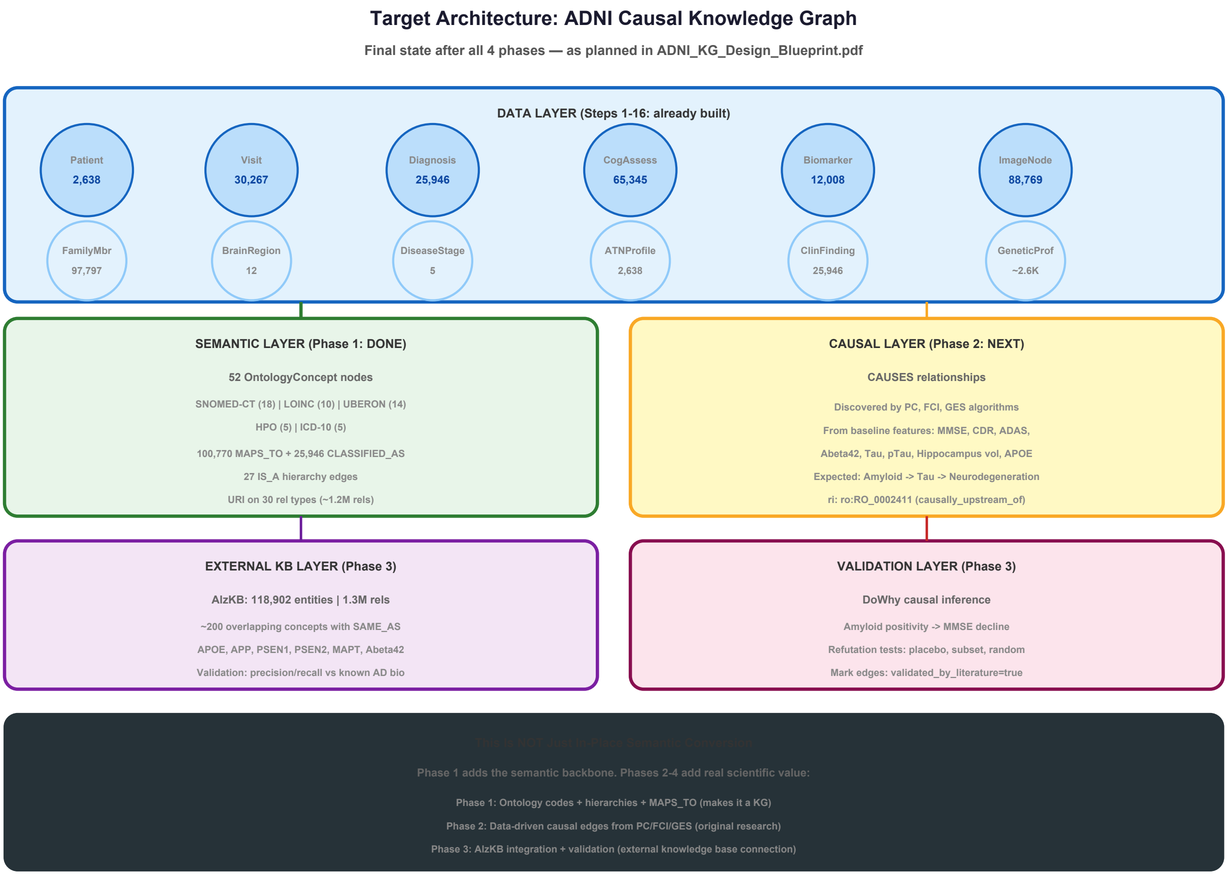

Figure 3: Target 4-layer architecture of the ADNI Causal Knowledge Graph.

- **Data Layer** (Steps 1-16, already built): 407,422 nodes from 108 ADNI tables. Patient-centric graph with visits, diagnoses, assessments, biomarkers, images, genetics, and family history. This is the foundation.
- **Semantic Layer** (Phase 1, DONE): 51 OntologyConcept nodes, SNOMED/LOINC/UBERON/HPO/ICD-10 codes on all data nodes, IS_A hierarchies, MAPS_TO and CLASSIFIED_AS edges, URI properties on all relationships. This makes it a Knowledge Graph.
- **Causal Layer** (Phase 2, NEXT): CAUSES relationships discovered by PC, FCI, and GES algorithms running on baseline patient features. Each CAUSES edge carries metadata: which algorithms found it, p-value, confidence, discovery date.
- **External KB + Validation Layer** (Phase 3): AlzKB integration via SAME_AS edges. Validation of causal edges against known AD biology. DoWhy causal inference for effect estimation. This layer connects our graph to the broader biomedical knowledge ecosystem.

8. Real Query Examples from Neo4j

These are **actual queries run against the live database**, showing what the semantic layer enables.

8.1 SNOMED-CT IS_A Hierarchy Traversal

Starting from Alzheimer's disease, we can traverse the full disease taxonomy:

```
Alzheimer's disease -> Dementia -> Neurodegenerative disorder -> Disorder of nervous system ->
Disease
```

Starting from MCI:

```
Mild cognitive impairment -> Neurodegenerative disorder -> Disorder of nervous system -> Disease
```

8.2 UBERON Brain Region Hierarchy

```
Hippocampal formation -> Cerebral cortex -> Brain
Entorhinal cortex -> Cerebral cortex -> Brain
Temporal lobe -> Cerebral cortex -> Brain
```

8.3 ICD-10 Classification Hierarchy

```
G30.9 (Alzheimer disease, unspecified) -> G30 (Alzheimer disease)
F06.7 (Mild cognitive disorder) -> F06 (Other mental disorders)
```

8.4 Full Cross-Domain Traversal

A single Cypher query can now traverse from a patient through clinical data to formal ontology concepts — something impossible before Phase 1:

```
Patient (002_S_0619) -[HAS_VISIT]-> Visit -[HAS_DIAGNOSIS]-> Diagnosis (AD)
  -[CLASSIFIED_AS]-> OntologyConcept (ICD-10: G30.9) -[IS_A]-> G30
  -[MAPS_TO]-> OntologyConcept (SNOMED: 26929004) -[IS_A]-> Dementia -[IS_A]-> Neurodegenerative
-[IS_A]-> Disorder of NS -[IS_A]-> Disease
```

9. Summary

- The graph has **407,422 nodes** and **1,391,812 relationships** from 108 ADNI tables.
- Phase 1 (completed today) transformed it from an LPG to a KG by adding ontology codes (SNOMED-CT, LOINC, UBERON, ICD-10) to all data nodes.
- **CN = Cognitively Normal** = healthy control. ICD-10 Z03.89 ('No diagnosis') is the correct code — it means 'examined and found healthy'. This is standard medical coding.
- The Knowledge Base connection is the **OntologyConcept layer** (51 nodes across 5 ontologies) with MAPS_TO, CLASSIFIED_AS, and IS_A relationships.
- Phase 1 is **not the end** — it is the semantic foundation for Phases 2-4 which will add causal discovery (PC/FCI/GES), AlzKB integration (external KB), and validation.
- The target architecture has **4 layers**: Data (done) + Semantic (done) + Causal (next) + External KB + Validation (planned).
- All relationship types carry **URI properties** from formal ontologies (Relation Ontology, SKOS, RDFS, OWL), making every edge machine-interpretable.

Report generated February 24, 2026. All node/relationship counts are live from Neo4j 5.24.2. Phase 1 complete. Next: Phase 2 (Steps 21-23: Causal Discovery).